

Instructions:

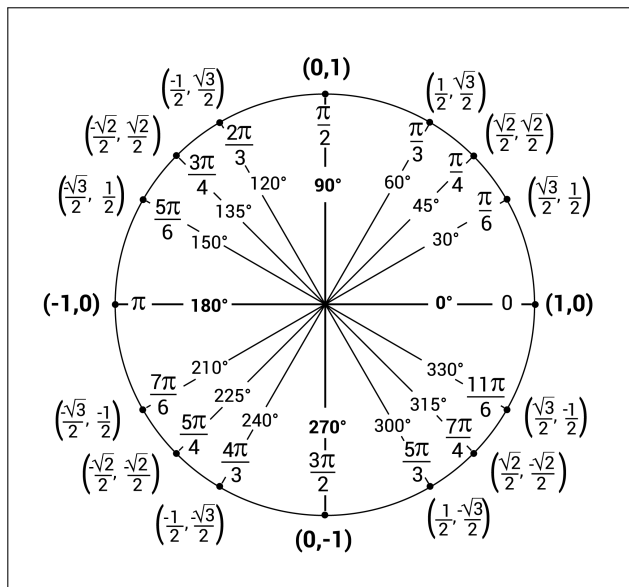
- Make sure to clearly show each step of your solution for full credit - correct answers without any work will not receive credit. Some questions may remind you to show your work explicitly, but you should assume each question requires you to justify your steps by making your thought process clear in your solution. You can use the solutions we write in class as an indication of how much work you are expected to show. You can ask the instructor if it is not clear how much work you are expected to show on a particular problem.
- Indicate your affirmation of the following statement by signing below.

I affirm that I did not give or receive any unauthorized assistance to solve the problems below. I understand that the penalty for academic dishonesty is an F for the course and possible suspension or expulsion from the university.

Signature: _____

Formulas

- $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1, \lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$
- Continuity at $x = a$: $\lim_{x \rightarrow a} f(x) = f(a)$
- $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- Basic derivatives
 - $\frac{d}{dx} (c \cdot f(x)) = c \cdot f'(x)$
 - $\frac{d}{dx} (f(x) \pm g(x)) = f'(x) \pm g'(x)$
 - $\frac{d}{dx} (x^n) = nx^{n-1}$
 - $\frac{d}{dx} (e^x) = e^x$
 - $\frac{d}{dx} (\ln(x)) = \frac{1}{x}$
 - $\frac{d}{dx} (\sin x) = \cos x$
 - $\frac{d}{dx} (\cos x) = -\sin x$
 - $\frac{d}{dx} (\tan x) = \sec^2 x$
 - $\frac{d}{dx} (\sec x) = \sec x \cdot \tan x$
 - $\frac{d}{dx} (\cot x) = -\csc^2 x$
 - $\frac{d}{dx} (\csc x) = -\csc x \cdot \cot x$
- Derivative rules
 - $\frac{d}{dx} (f(x) \cdot g(x)) = f'(x) \cdot g(x) + g'(x) \cdot f(x)$
 - $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$
 - $\frac{d}{dx} (f(g(x))) = f'(g(x)) \cdot g'(x)$
- $L(x) = f'(a)(x - a) + f(a)$
- MVT: $f'(c) = \frac{f(b) - f(a)}{b - a}$



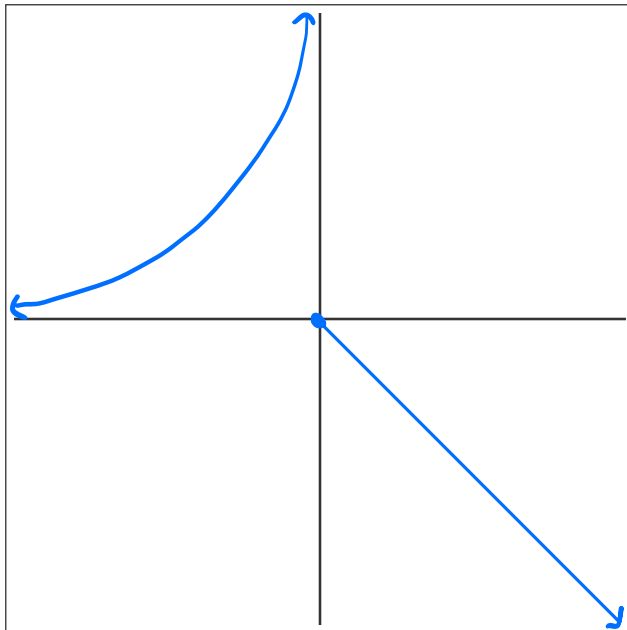
A1 - Limits and continuity (inspired by Brightspace problems 1, 4, and 6 and part 1 of the class notes)

- (a) Use the axes to the right to sketch the graph of one function $f(x)$ that satisfies all of the following conditions.

$$\lim_{x \rightarrow -\infty} f(x) = 0 \quad \lim_{x \rightarrow 0^-} f(x) = \infty$$

$$\lim_{x \rightarrow 0^+} f(x) = 0 \quad \lim_{x \rightarrow \infty} f(x) = -\infty$$

one possible answer →



- (b) Find each limit or show that it doesn't exist. Clearly show each step of your solution for full credit.

(i) $\lim_{x \rightarrow 0^+} \sqrt{x} \left[1 + \sin^2 \left(\frac{2\pi}{x} \right) \right]$

Hint: $0 \leq \sin^2 \left(\frac{2\pi}{x} \right) \leq 1$

$$1 \leq 1 + \sin^2 \left(\frac{2\pi}{x} \right) \leq 2$$

$$\sqrt{x} \leq \sqrt{x} \left(1 + \sin^2 \left(\frac{2\pi}{x} \right) \right) \leq 2\sqrt{x}$$

$x \rightarrow 0^+$ ↓

↓ $x \rightarrow 0^+$

$$0 \leq \lim_{x \rightarrow 0^+} \sqrt{x} \left(1 + \sin^2 \left(\frac{2\pi}{x} \right) \right) \leq 0$$

||
0

(By Squeeze Theorem)

(ii) $\lim_{x \rightarrow -6} \frac{2x + 12}{|x + 6|}$

$$\lim_{x \rightarrow -6^-} \frac{2(x+6)}{-(x+6)} = -2$$

$$\lim_{x \rightarrow -6^+} \frac{2(x+6)}{(x+6)} = 2$$

→ DNE

- (c) For what value of the constant c is the function $g(x)$ continuous at $x = 2$?

$$g(x) = \begin{cases} cx^2 + 2x & \text{if } x < 2 \\ x^3 - cx & \text{if } x \geq 2 \end{cases}$$

$$\lim_{x \rightarrow 2^-} g(x) = c \cdot 2^2 + 2 \cdot 2 = 4c + 4$$

$$\lim_{x \rightarrow 2^+} g(x) = g(2) = 2^3 - c \cdot 2 = 8 - 2c$$

$$4c + 4 = 8 - 2c \rightarrow$$

$$\boxed{c = \frac{2}{3}}$$

A2 - The derivative (inspired by part 2 of the class notes and Brightspace problem 1)

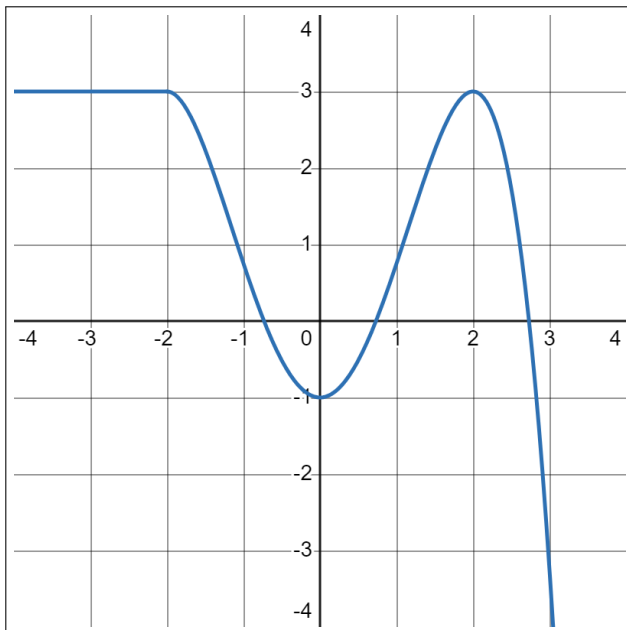
(a) The graph of a function $f(x)$ is shown to the right.

(i) List five different integers x for which $f'(x) = 0$.

$$x = -4, -3, -2, 0, 2$$

(ii) List two different integers x for which $f'(x)$ is negative.

$$x = -1, 3$$



(b) Use the limit definition of the derivative to determine if the piecewise function

$$f(x) = \begin{cases} -2x - 9 & x < -3 \\ x^2 + 4x + 4 & x \geq -3 \end{cases}$$

is differentiable at $x = -3$. For full credit, you must use the limit definition of the derivative. Clearly show all of your work.

$$\lim_{h \rightarrow 0^-} \frac{f(-3+h) - f(-3)}{h} = \lim_{h \rightarrow 0^-} \frac{-2(-3+h) - 9 - ((-3)^2 + 4(-3) + 4)}{h} =$$

$$\lim_{h \rightarrow 0^-} \frac{6 - 2h - 9 - (9 - 12 + 4)}{h} = \lim_{h \rightarrow 0^-} \frac{-4 - 2h}{h} = \infty$$

$$\lim_{h \rightarrow 0^+} \frac{f(-3+h) - f(-3)}{h} = \lim_{h \rightarrow 0^+} \frac{(-3+h)^2 + 4(-3+h) + 4 - 1}{h} = \lim_{h \rightarrow 0^+} \frac{h^2 - 2h}{h} = -2$$

$-2 \neq \infty$ so $f'(-3)$ does not exist.

A3 - Derivative rules (inspired by past final worksheet problem 3 and part 2 of the class notes)

Find the derivative of each of the following functions. You do not need to simplify your answers.

(a) $f(x) = x^{\ln(x)}$

$$\ln(f(x)) = \ln(x) \cdot \ln(x) = [\ln(x)]^2$$

$$\frac{f'(x)}{f(x)} = \left(2 \cdot \ln(x) \cdot \frac{1}{x} \right)$$

$$\rightarrow \boxed{f'(x) = x^{\ln(x)} \cdot \left(\frac{2 \ln(x)}{x} \right)}$$

(b) $g(x) = \frac{e^x(x^2 + 1)}{(\tan x + \cos x)^3}$

$$g'(x) = \frac{(\tan x + \cos x)^3 \cdot (e^x \cdot (x^2 + 1) + e^x \cdot 2x) - e^x(x^2 + 1) \cdot 3(\tan x + \cos x)^2 \cdot (\sec^2 x - \sin x)}{(\tan x + \cos x)^6}$$

A4 - Implicit curves and linearization (inspired by Brightspace problems 6 and 7 and past final worksheet problem 4)

- (a) Find a point on the curve $2y^5e^x = 3y - 6x$ at which the tangent line to the curve has a slope equal to 2, or show that no such point exists.

$$\begin{aligned}
 2y^5e^x + 10y^4 \frac{dy}{dx} e^x &= 3 \frac{dy}{dx} - 6 \\
 2y^5e^x + 10y^4 \cdot 2e^x &= 3 \cdot 2 - 6 \\
 2y^4e^x(y+10) &= 0 \rightarrow \begin{matrix} y=0 \\ y=10 \end{matrix}
 \end{aligned}$$

$y=0 \rightarrow x=6$
 So $(0,0)$ is such a point.
 $(y=10 \rightarrow x=?)$

- (b) Let $f(x) = \sqrt[3]{28 - x^2}$.

- (i) Find the linear approximation of $f(x)$ at $x = 1$.

$$\begin{aligned}
 f'(x) &= \frac{1}{3} (28 - x^2)^{-\frac{2}{3}} \cdot -2x \\
 f'(1) &= \frac{1}{3} (28 - 1)^{-\frac{2}{3}} \cdot -2 \cdot 1 = \frac{-2}{27} \quad L(x) = \frac{-2}{27} (x-1) + 3 \\
 f(1) &= \sqrt[3]{28 - 1^2} = 3
 \end{aligned}$$

- (ii) Use your answer from part (i) to approximate $\sqrt[3]{27.19}$. You do not need to simplify your answer.

$$\begin{aligned}
 \sqrt[3]{27.19} &= \sqrt[3]{28 - 0.81} = f(0.9) \approx \frac{-2}{27} (0.9 - 1) + 3 \\
 &= \frac{1}{135} + 3
 \end{aligned}$$

A5 - Related rates (inspired by Brightspace problem 2 and the class notes)

- (a) The radius of a sphere is increasing at a rate of 4 mm/s. How fast is the volume increasing when the diameter is 80 mm? You don't need to simplify your answer, but you should include units. The volume of a sphere with radius r is

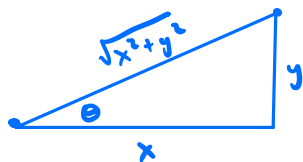
$$V = \frac{4}{3}\pi r^3$$

$$\frac{dv}{dt} = \frac{4}{3}\pi \cdot 3r^2 \cdot \frac{dr}{dt}$$

$$d=80 \rightarrow r=40$$

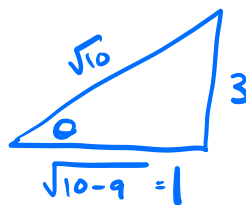
$$\frac{dv}{dt} = \frac{4}{3}\pi \cdot 3(40)^2 \cdot 4 = 25600\pi \text{ mm}^3/\text{sec}$$

- (b) A plane flies over an observer in such a way that the horizontal distance from the observer to the point on the ground directly below the plane is increasing at a rate of 400 miles per hour, and the altitude (height) of the plane is increasing at a rate of 30 miles per hour. Find the rate at which the angle of elevation from the observer to the airplane is changing when the plane is 3 miles above the ground and the straight line distance from the observer to the plane is $\sqrt{10}$ miles. You don't need to simplify your answer, but you should include units.



$$\frac{dx}{dt} = 400$$

$$\frac{dy}{dt} = 30$$



$$\begin{aligned} x &= 1 \\ y &= 3 \\ x^2 + y^2 &= 10 \\ \cos^2 \theta &= \left(\frac{1}{\sqrt{10}}\right)^2 = \frac{1}{10} \end{aligned}$$

$$\tan \theta = \frac{y}{x}$$

$$\sec^2 \theta \cdot \frac{d\theta}{dt} = \frac{x \cdot \frac{dy}{dt} - y \cdot \frac{dx}{dt}}{x^2}$$

$$\begin{aligned} \frac{d\theta}{dt} &= \frac{x \cdot \frac{dy}{dt} - y \cdot \frac{dx}{dt}}{x^2} \cdot \cos^2 \theta = \frac{1 \cdot 30 - 3 \cdot 400}{1^2} \cdot \frac{1}{10} \\ &= -117 \text{ rad/hr} \end{aligned}$$

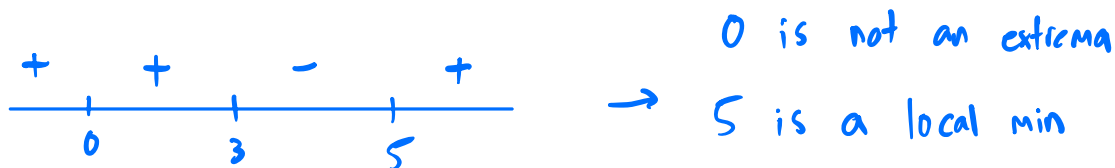
B1 - Extrema of functions (inspired by Brightspace problem 9 and past final worksheet problem 1)

Consider the function $f(x) = x^3(x - 5)^2$. The first and second derivatives of $f(x)$ are given below.

$$f'(x) = 5x^2(x - 3)(x - 5)$$

$$f''(x) = 10x(2x^2 - 12x + 15)$$

- (a) What does the first derivative test say about the critical points $x = 0$ and $x = 5$?



- (b) What does the second derivative test say about the critical points $x = 0$ and $x = 5$?

$$f''(0) = 0$$

inconclusive at 0

$$f''(5) = 50 \cdot (50 - 60 + 15) = 250 > 0$$

→ 5 is a local min

- (c) Find the absolute minimum and absolute maximum of $f(x)$ on the interval $[0, 4]$. Report your answers as ordered pairs.

$$f(0) = 0$$

$$f(3) = 3^3 \cdot (-2)^2 = 108$$

$$f(4) = 4^3 \cdot (-1)^2 = 64$$

→

$(0, 0)$ is the absolute min

$(3, 108)$ is the absolute max

B2 - The Mean Value Theorem (inspired by past final worksheet problem 2 and the class notes)

Consider the function $f(x) = 1 - x^2 + \frac{1}{100}e^{2x}$.

- (a) Show that there are at least two points where the slope of $f(x)$ is equal to the value of $f(x)$, i.e., where $f'(x) = f(x)$.

$$f'(x) = -2x + \frac{1}{50}e^{2x}$$
$$-2x + \frac{1}{50}e^{2x} = 1 - x^2 + \frac{1}{100}e^{2x} \rightarrow \underbrace{x^2 - 2x - 1 + \frac{1}{100}e^{2x}}_{g(x)} = 0$$

$g(x)$ is continuous, so we use IVT.

$$\left. \begin{aligned} g(-1) &= (-1)^2 - 2(-1) - 1 + \frac{1}{100}e^{-2} = 2 + \frac{1}{100}e^{-2} > 0 \\ g(0) &= 0^2 - 2 \cdot 0 - 1 + \frac{1}{100}e^0 = -1 + \frac{1}{100} < 0 \\ g(3) &= 3^2 - 2 \cdot 3 - 1 + \frac{1}{100}e^6 = 2 + \frac{1}{100}e^6 > 0 \end{aligned} \right\} \begin{array}{l} \text{There is a solution} \\ \text{in } (-1, 0) \text{ and} \\ (0, 3), \text{ by IVT.} \end{array}$$

- (b) Show that there are at most two points where the slope of $f(x)$ is equal to the value of $f(x)$.

$g(x)$ and $g'(x)$ are differentiable, so we apply the corollary to Rolle's Theorem.

$$g'(x) = 2x - 2 + \frac{1}{50}e^{2x}$$

$$g''(x) = 2 + \frac{1}{25}e^{2x} > 0 \text{ for all } x, \text{ so } g''(x) \text{ has 0 roots.}$$

By the corollary, $g'(x)$ has at most 1 root, so $g(x)$ has at most two roots.

B3 - Curve sketching (inspired by the class notes)

Suppose a function $f(x)$ has the following first and second derivatives:

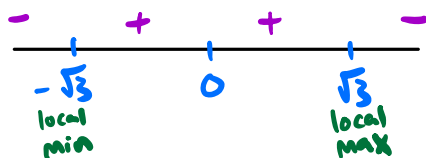
$$f'(x) = \frac{-x^2 + 3}{x^4}$$

$$f''(x) = \frac{2x^2 - 12}{x^5}$$

- (a) What are the x -values of the local maxima/minima of f (if any)? Classify each as a maximum or minimum.

$$f'(x) = 0 \rightarrow x^2 = 3 \rightarrow x = \pm\sqrt{3}$$

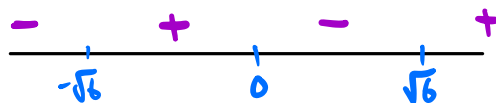
$$f'(x) \text{ DNE @ } x=0$$



- (b) What are the x -values of the points of inflection of f (if any)?

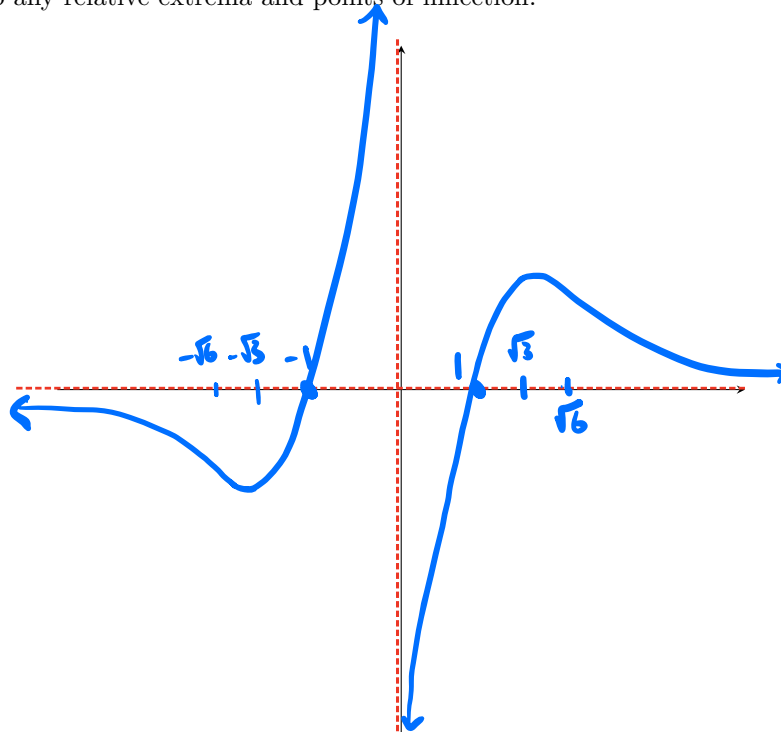
$$f''(x) = 0 \rightarrow 2x^2 - 12 = 0 \rightarrow x = \pm\sqrt{6}$$

$$f''(x) \text{ DNE @ } x=0$$



$\pm\sqrt{6}$ are points of inflection

- (c) Suppose we know that $f(x)$ has a vertical asymptote at $x = 0$, a horizontal asymptote at $y = 0$, and x -intercepts at $x = 1$ and $x = -1$. Sketch a graph of f below and label the points on the x -axis corresponding to any relative extrema and points of inflection.



B4 - Optimization (inspired by Brightspace problem 2 and part 3 of the class notes)

- (a) What is the minimum vertical distance between the parabolas $y = -8x^2 + 9x - 3$ and $y = -4x^2 + 17x + 6$.

$$D(x) = (-4x^2 + 17x + 6) - (-8x^2 + 9x - 3) = 4x^2 + 8x + 9$$

$$D'(x) = 8x + 8 = 0 \rightarrow x = -1$$

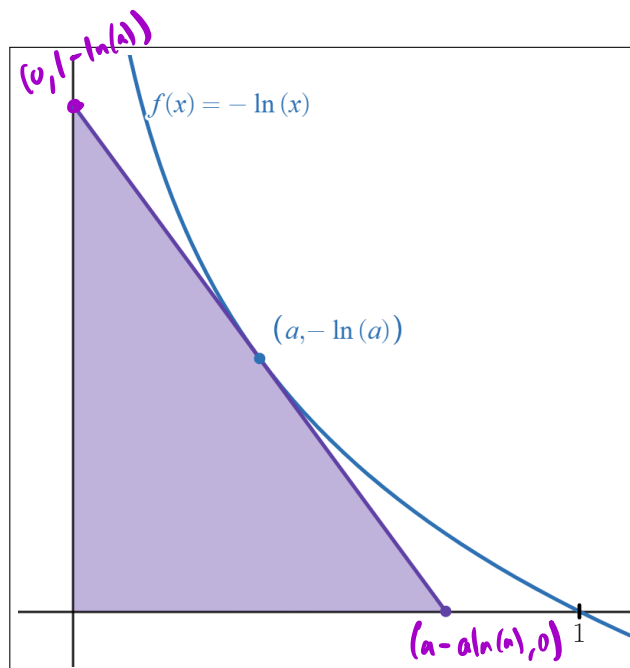
$$D(-1) = 4(-1)^2 + 8(-1) + 9 = \boxed{5}$$

- (b) A right triangle is formed from the x and y axes along with a line tangent to $f(x) = -\ln(x)$ at a point $x = a$ as shown in the picture. Use the following steps to find the value of a that maximizes the area of this triangle.

Step 1: Find an expression for the line tangent to $f(x)$ at some point a . Since a is unspecified, the expression for this line will depend on a .

Step 2: Find expressions for the x and y intercepts of the line from Step 1. These intercepts will also be in terms of a .

Step 3: The intercepts from Step 2 are the width and height of the triangle, so use them to express the area in terms of a . Now use calculus to find the value of a that maximizes this area (a should be between 0 and 1 because $f(x)$ has an x intercept of 1 as shown in the picture).



$$\text{Step 1: } L(x) = f'(a)(x-a) + f(a) = -\frac{1}{a} \cdot (x-a) - \ln(a)$$

$$\text{Step 2: } y\text{-int: } L(0) = -\frac{1}{a} \cdot (0-a) - \ln(a) = 1 - \ln(a)$$

$$x\text{-int: } 0 = -\frac{1}{a}(x-a) - \ln(a) \rightarrow x = a - a \ln(a)$$

$$\text{Step 3: } \text{Area} = \frac{1}{2} \cdot (a - a \ln(a)) \cdot (1 - \ln(a)) = \frac{1}{2} (a - a \ln(a) - a \ln(a) + a \ln(a)^2)$$

$$= \frac{1}{2} a - a \ln(a) + \frac{1}{2} a \ln(a)^2$$

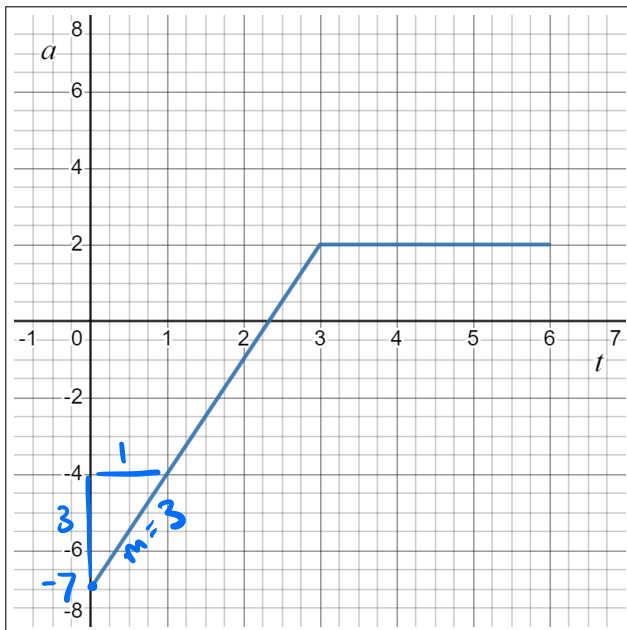
$$\text{Area}' = \frac{1}{2} - (a \cdot \frac{1}{a} + \ln(a)) + \frac{1}{2} (a \cdot 2 \ln(a) \cdot \frac{1}{a} + \ln(a)^2) = \frac{1}{2} - 1 - \cancel{\ln(a)} + \cancel{\ln(a)} + \frac{1}{2} \ln(a)^2 = 0$$

$$\rightarrow 0 = -\frac{1}{2} + \frac{1}{2} \ln(a)^2 \rightarrow \ln(a) = \pm 1 \rightarrow a = e^1 \text{ or } \boxed{e^{-1}}$$

↑
in $[0, 1]$

B5 - Antiderivatives (inspired by Brightspace problem 10 and part 2 of the class notes)

An object's acceleration (in m/s^2) at t seconds is given by the function $a(t)$ which is graphed to the right. Suppose the object's velocity at $t = 1$ is $3 m/s$.



- (a) Find the object's velocity as a function of time, $v(t)$, from $t = 0$ to $t = 3$.

$$a(t) = 3t - 7$$

$$v(t) = \frac{3}{2}t^2 - 7t + C$$

$$v(1) = \frac{3}{2} \cdot 1^2 - 7 \cdot 1 + C = 3$$

$$\rightarrow C = 8.5$$

$$v(t) = 1.5t^2 - 7t + 8.5$$

- (b) Find the object's change in position from $t = 0$ to $t = 3$. You don't need to simplify your answer.

$$s(t) = 0.5t^3 - 3.5t^2 + 8.5t + D$$

$$s(3) - s(0) = 0.5 \cdot 3^3 - 3.5 \cdot 3^2 + 8.5 \cdot 3 + D - (D)$$

$$= 0.5 \cdot 3^3 - 3.5 \cdot 3^2 + 8.5 \cdot 3$$

- (c) Find the object's velocity as a function of time, $v(t)$, from $t = 3$ to $t = 6$.

$$a(t) = 2$$

$$v(t) = 2t + A$$

$$v(3) = \underbrace{1.5 \cdot 3^2 - 7 \cdot 3 + 8.5}_{\text{from part (a)}} = 1$$

$$v(3) = 2 \cdot 3 + A = 1 \rightarrow A = -5$$

$$v(t) = 2t - 5$$

- (d) Find the object's position as a function of time, $s(t)$, from $t = 3$ to $t = 6$ if $s(3) = 0$.

$$s(t) = t^2 - 5t + B$$

$$s(3) = 3^2 - 5 \cdot 3 + B = 0 \rightarrow B = 6$$

$$s(t) = t^2 - 5t + 6$$